PE&RC PhD PROJECT PROPOSAL

*Please submit the signed PDF of the PE&RC PhD Project Proposal by email to the PE&RC office (office.pe@wur.nl) no later than **6 months** after the start of the PhD project. In case a peer-reviewed full project proposal is available (e.g., NWO or EU), please send that proposal along together with the reviewers' comments, rebuttal(s) and the final letter of acceptance.*

Part 1: General information

1.1 ABOUT THE PROJECT PROPOSAL AND EVALUATION CRITERIA:
The PE&RC PhD proposal is written by the PhD candidate in consultation with and feedback from the supervisory team. After submission, the proposal is sent out for review. The candidate and the supervisory team are encouraged to suggest possible independent reviewers that can provide constructive feedback to the candidate.

The reviewers are asked to evaluate the proposal on:

- *Scientific quality*
  Clarity of writing, depth of literature review, definition of research question(s) and objective(s), and originality of subject and/or methodology.
- *Feasibility*
  Feasibility given the 4-year time frame, (financial) resources, effectiveness of the proposed approach/strategy, logistical/practical aspects and evaluation of the expertise within the supervising team and/or collaborations for adequate support in the proposed project.
- *Scientific and/or societal impact*
  The importance of the potential research results in the short and long term in the field of research, potential use and relevance in other scientific disciplines, and potential use and relevance for society (economic, technical, social, cultural), including technological application.

1.2 GENERAL PROJECT INFORMATION:

Main PE&RC affiliated Institute/University	Wageningen University & Research
Main PE&RC research group	Soil Geography & Landscapes group
Other PE&RC research groups involved	-
Where will the research be conducted (country)	Netherlands & Nepal
Full name of the PhD candidate	Zoë Sophia Kleijwegt
Project duration	FROM 01/09/2022 TO 01/09/2026
Time spent on PhD project	☑ Fulltime ☐ Part-time: for ... hours per week
Principal supervisor/Promotor	Prof. Dr. Jakob Wallinga
Additional supervisor(s)	Dr. Annegret Larsen, Dr. Jeroen M. Schoorl

1.3 ETHICS:
Ethical issues such as experiments with animals (vertebrates), work with genetically modified organisms, work with material under the Nagoya protocol, or personal data privacy considerations (AVG) may have legal implications. Also, ethical guidelines of the respective research institution may apply.

Do any of the above mentioned, or other ethical issues apply to this project?	~~YES~~ / **NO**
If YES, please elaborate:	

1.4 PE&RC RESEARCH THEME(S)

In which of the PE&RC research theme(s) does the project fit (multiple options possible). Please elaborate.

Themes	**Elaboration**
☑ Biodiversity and Systems Dynamics ☐ One Health ☐ Redesign of (argo-) ecological systems ☐ Data and Engineering Science for Life and Environment	*The project aims to understand the landscape dynamics of Bardia National Park in order to optimally manage & maintain grasslands, groundwater, and Bengal tiger populations.*

Part 2: Proposal structure

2.1 Title

Providing a long-term perspective on the landscape dynamics of tiger habitat in the Terai Arc Landscape by combining geomorphological and paleoecological reconstruction.

2.2 Abstract

Large predators such as the continental tiger *(Panthera tigris tigris)* are particularly sensitive to the global biodiversity crisis. The Terai Arc Landscape (TAL) is one of the largest continuous habitats for this species, and due to conservation efforts, the population has been recovering significantly in this area. The increased numbers are however widely resulting in human wildlife conflict, raising questions about how to maintain a viable population and a sustainable habitat. This PhD project therefore aims to increase understanding of the habitat by reconstructing the past TAL landscape from the perspective of the tiger, through establishing the geomorphological-, land use- and animal history. These histories will be reconstructed using a mix of luminescence dating, pollen, phytolith-, charcoal- and sedimentary ancient DNA analysis. Based on this reconstruction, environmental baselines of the tiger habitat will be established. These baselines can then be used by conservationists to establish optimal management practices, to achieve a sustainable tiger habitat with a realistic and viable tiger population.

2.3 Topic & Background

The global threatening of biodiversity affects ecosystems extensively (Rands et al., 2010). Apex predators play a crucial role in maintaining healthy landscapes (Laundré et al., 2010), but their prevalence in human-wildlife conflicts makes them particularly vulnerable in the biodiversity crisis. A prime example of such vulnerability is the continental tiger (*Panthera tigris tigris*). Over the past hundred years, the population dramatically decreased due to poaching, hunting, and habitat encroachment, from an estimated 100,000 in the early 1900s to only 3,000 in 2010 (WWF, 2011). In recent years, extensive efforts have been made to revive the population, and one of the main targeted areas for the conservation of this species is the Terai Arc Landscape (TAL) (Gour & Reddy, 2015). Located at the foothills of the Himalaya Mountain range in Nepal and India, it consists of fourteen protected areas containing wetlands, grasslands, and forests. The area forms not only an important habitat for the continental tiger, but for other iconic species such as the Indian Elephant (*Elephas maximus indicu*s), Greater One Horned Rhino (*Rhinocerus unicornis*) and an enormous range of bird- and plant species, making it an important biodiversity hotspot.

Bardia National Park (BNP) is a core tiger habitat within the TAL (Nepali Times, 2018), located in the southwest of Nepal. The tiger population in Nepal has been increasing since 2010, when the Tx2 initiative was founded, aiming to double the Tiger population by 2022. This goal was reached in Nepal by growing the population from 121 in 2009 to 235 in 2017 (DNPWC & DFSC, 2018), and is now estimated at 355 (DNPWC & DFSC, 2022). The increase in population was achieved through the reduction of poaching and improving the protection of national parks, but due to increased human inhabitance of the landscape since the dwindling of the tiger population, the current growth of the tiger population is causing a rise in human-wildlife conflict. This conflict is caused by tigers that attack people (man-eaters), as well as the Indian leopard (*Panthera pardus fusca*), which is known to move into the villages under habitat encroachment (Madhusudan & Karanth, 2002). The human-wildlife conflict threatens the tiger population its viability as it increases the risk of violence towards the tiger (DNPWC, 2016), so ideal management practices for areas considered tiger habitat are still sought.

The history of Bardia National Park

Currently, little is known about the development of the landscape surrounding BNP. The park is located directly east of a Duar ("door") where the Karnali river exits the Himalayan foothills and enters the open plains of the Terai, forming a mega fan from the Duar to approximately 40 km south (see Fig. 2). The activity of this fan may well have contributed to the formation of the landscape now known as Bardia National Park, by affecting the geomorphology and the groundwater levels throughout the park. This would affect the suitability for grasslands

to develop, resulting in tiger habitat changes. Another reason that BNP and the fan are of particular interest, is that the inherent openness of landscape that an active river fan provides could have been of interest for humans. The riverine grasslands would have been easily accessible for i.e., cattle grazing, and archaeological data from nearby sites such as the Deokhuri Dun Valley (Corvinus, 2007), show signs of human migration starting in the Paleolithic, with most finds originating from the Mesolithic period, but Neolithic finds were minimal. The general population in the Terai is widely thought to have remained small due to the high prevalence of *Malaria tropica*. The indigenous *Tharu* community appears resistant to the disease, allowing for them to have maintained small settlements in the area. Malaria eradication in the 1960s caused a surge in immigration to the fertile agricultural lands, resulting in the high population density seen today. The human history prior to this surge, and the effects they may have had on the landscape are however largely unknown on a centennial to millennial scale (Miehe et al., 2015), and the recent anthropogenic effect on the landscape is therefore yet to be determined.

The Tharu from Chitwan (ca. 250 km east of Bardia) have a traditional land use practice that is cereal based (i.e. wheat, rice). Low intensity shifting cultivation was practiced, exchanging rice and mustard plantations, until this practice was outlawed in 1951 CE. In support of these agricultural practices, the Tharu have a longstanding history of keeping livestock for both livelihood and cultural purposes. On the Tharu from the Bardia region specifically, literature is very limited. It is understood that the Tharu migrated from the south, based on their Indo-Aryan lineage, but the exact timing of settlement is unknown. Chitwan, however, has traces of Tharu settlements tracing back to circa 1500 CE, so it is thought that the occupation in Bardia spans at least several centuries (Müller-Boker, 1999). General patterns of human occupation that could be observed in the vegetation record are a decrease of forest cover and an increase of grass- and herbaceous species, alongside the appearance of cultivar species such as rice, possibly maize and certain *Brassicaceae* species (i.e. mustard). It is also likely that an increase in fire frequency would occur, although the Terai Arc Landscape does burn naturally, so interpretations of fire as a human indicator should be made carefully, especially at times of drought.

Nepal falls under influence of the Indian summer monsoon, and the general observation is that monsoon intensity has decreased throughout the Holocene as a result of orbital forcing (i.e. , Fleitmann et al., 2007, Dong et al., 2010). Broad climate patterns are available for the area, although most paleorecords have been collected in the higher mountainous regions of Nepal. To estimate the most relevant general climatological patterns, we here describe the findings of the record from lake Rara, located ca. 100 km north-east from Bardia National Park. Mainly, an approximation of wind strength variability was made to increase the understanding of monsoonal patterns at a 70-year resolution. The regional patterns corresponds with the decrease of monsoon intensity throughout the Holocene, with one weak-wind event captured within our timeframe of interest, making it likely we could record a drier period around 500 CE (Nakamura et al., 2012), which should be weighed in our interpretations.

Bardia National Park as a Tiger refuge

To determine how the habitat of the tiger in Nepal can best be developed, managed, and maintained under increasing anthropogenic influence, it is important to build an understanding of what the ideal tiger habitat is, and how this habitat developed over time. Therefore, in 2020, the *Save the Tiger, Save the Water, Save the Grasslands!* Project was initiated and awarded funding from the Dutch Research Council (NWO). Within this consortium, the aim is to develop sustainable tiger habitat for a viable tiger population in the TAL (Griffioen et al., 2019). Work package 5, titled “long-term interactions between river dynamics, vegetation, groundwater, land-use and animals” encompasses this PhD-project and aims to provide a **long-term perspective on the landscape dynamics of tiger habitat in the Terai Arc Landscape through combining of geomorphological and paleoecological reconstruction**. This reconstruction can then be used to identify which factors drive the landscape, and what needs to be considered when establishing baselines for the habitat of the continental tiger.

The focus of this project lies in understanding transitions in quantity and location of forests and grasslands, as it is widely understood that this shapes the suitability of the habitat for herbivores and their predators. Increasing

the availability grass (a high value energy source) is thought to increase deer populations which in turn sustains a larger tiger population. To better understand how the Karnali river fan dynamics changed and affected the landscape, forest and grasslands over time, particular interest lies in the past size and position of the river fan, as a wider or narrower fan would affect the ratio of riverine grasslands to forest in the region. In addition, if there was higher channel activity in recent times (Holocene) in the Bardia National Park forest, this would affect the current successional phase of BNP and therefore the state of the forest. By identifying former channel belts and meanders, and determining when they were last active through luminescence dating, these morphological characteristics of BNP and the Karnali River fan and surrounding channels can be better understood. In addition to the riverine grasslands however, BNP also contains several phantas, which are savanna-like grasslands away from the river. These grasslands are heavily maintained by fire and are thought to have originated from human activity rather than natural development as well. Therefore, distinguishing between these two types from different origins is important to understand the development of the landscape and the effects the different types of grassland have on tiger habitat suitability.

To establish past landscape dynamics as a whole, changes in vegetation, land use and animal presence over time need to be quantified. To achieve this, paleoecological analysis can potentially provide a long-term temporal record. Currently, there are no paleoecological records of the Nepalese western Terai region, so this project aims to generate said data. Generally speaking, BNP consists of a mixture of Sal forest (*Shorea robusta*), riverine grasslands and phantas. The composition of these vegetation assemblages is not currently known in detail, but is being studied within the consortium. This data will be available to aid the interpretation of paleoecological records.

The grassland-to-forest ratio that we are interested in, in regards to tiger habitat suitability, can be calculated from phytoliths and pollen on a local and a regional resolution, respectively (Strömberg et al., 2007). The phytoliths of grasses also differ in morphology and assemblages between species, so transitions in ratios of riverine grassland (dominated by *Saccharum bengalensis*) to phantas (dominated by *Imperata cylindrica*) may be captured by this proxy. Additionally, phytoliths can indicate the local cultivation of major crops such as maize and rice (Piperno, 2006), which can help to pinpoint the onset and extent of agricultural land use practices. Coprophilous fungal spores can also provide insight on this, as they are a proxy for herbivore density and will therefore reflect relative livestock abundance. Finally, sedimentary ancient DNA (sedaDNA) analysis (Thomsen & Willerslev, 2015) can be applied to determine if the tiger itself was present in the system over time, and how this presence responded to landscape changes.

Combining paleoecological records with the outcome of the geomorphological analysis will provide a chronology to increase understanding of the evolution of the landscape, and how this evolution was driven. If the fan indeed varied in surface area over time, it is expected that riverine grasslands are more prevalent at times of more channel activity. This may increase prey availability and therefore the presence of the tiger. Anthropogenic activity may also affect tiger presence. The appearance of people is known to correspond with an increase in fire activity and deforestation which may increase grassland cover but could result in more phantas rather than riverine grasslands. How this distinction affects the tiger (and prey) presence is, however, unknown. Finally, the appearance of livestock may also affect the availability of prey and affect the tiger in that way. By providing a full reconstruction, transitions in the landscape and different states (baselines) can be identified, which can be used as benchmarks to compare the current state of the park (quantified by other work packages) to. Understanding the differences between the past and current system and the outcomes these differences have for the tiger habitat can guide conservation policies.

2.4 Scientific & societal impact

The output of this PhD project entails a reconstruction of the past landscape, using cutting edge technologies and combining these with established methodologies. This should provide important, reliable scientific insights that can be used by other members of the consortium to support and understand their own findings. The past

responses of wildlife to environmental change will feed into the understanding of the current dynamics of vegetation-prey-tiger dynamics, while the past responses of vegetation to the geomorphological and climatic changes will support the understanding of the current morphological development and ecohydrology of the system. Together, this will guide the development of conservation practices in the area.

Additionally, scientific impact can be found in the further development of methodologies. Specifically, sedaDNA analysis will be applied to a sub-tropical research area to track mammal presence. Most analysis of mammal sedaDNA thus far has focused on records from glacial lakes, as preservation under these conditions is favorable (Jia et al., 2022). However, it was found that the preservation is not necessarily driven climate, and preservation in the sub-tropics was studied and there is evidence to support that it is possible to extract mammal sedaDNA from (sub-)tropical settings as well (Bremond et al., 2017). This PhD-project will put this finding to the test by applying sedaDNA analysis targeted at the continental tiger to sub-tropical lacustrine sediment records.

Luminescence dating techniques were applied to young Himalayan deposits on several occasions, with limited success. Dingle et al. (2020) were able to generate ages from river sediment samples using OSL (Optically Stimulated Luminescence) at the Karnali river fan, but the ages show notably high uncertainties ranging from 10-50%. Additionally, samples were collected mostly in the north-west of the river fan, while the core tiger habitat is located on the south eastern side. This PhD-project will further explore the challenges of dating young Himalayan sediments with support of the excellent facilities and researchers at the Netherlands Centre for Luminescence dating, while aiming to increase the spatial coverage of luminescence dates at the Karnali river fan to improve the understanding of local landscape evolution that is relevant for the tiger habitat.

Generally speaking, the integration of the wide range of methodologies presented here can be applied in a wide context if proven successful. It has the potential to revolutionize the approach of habitat conservation, where conservation benchmarks can be based on much more than the known circumstances of recent times only.

In terms of societal impact, through the consortium, this PhD-project contributes to establishing a sustainable tiger habitat for a viable tiger population. This has the potential to improve the quality of life of the local communities in various ways. Firstly, ensuring that enough, healthy habitat is available for tigers and leopards can reduce human-wildlife conflict, mitigating the risks of injury, death, and monetary loss. Additionally, a landscape with a healthy population of one of the main charismatic wildlife species can increase tourism revenues. Having a perspective of this species its past existence in the area can further support this increase. Finally, sharing information gained from this project with the local community will improve the understanding and knowledge of the landscape, to inform best management practices both in and outside the national park area.

2.5 Detailed description of the research plan

The project will be structured in three chapters, followed by an integrative discussion and conclusion. The first chapter is centred around the geomorphological reconstruction of the landscape, the second chapter aims to uncover past vegetation patterns & land use and the third chapter will focus on tracking changes in the animal presence over time. These three chapters together feed into the synthesis of a mixed methodological landscape reconstruction that can provide environmental baselines of the tiger habitat in the Terai Arc Landscape.

At the start of the project, in November 2022, a field visit was performed to scope the research area for potential locations for both palaeoecological and geomorphological sampling. The visit was prepared by extensively investigating satellite imagery of the area to find potential sites of interest. In the field, the forest of BNP and the fan between the Kauriala and the Geruwa branch were explored by car and foot, with the support and guidance of experienced local park rangers who provided extensive information on the area. The insights from this visit heavily informed the selection of the sampling sites. In the following sections, the selections of these sites and the planned methodologies are described per chapter.

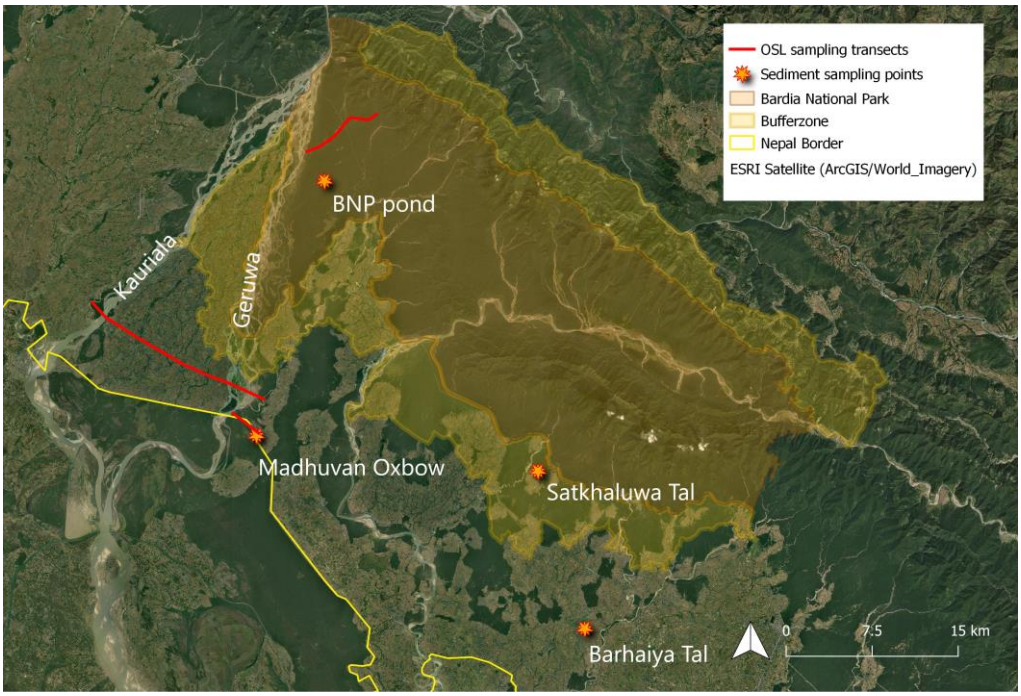

Figure 1. Overview of the research area, showing the Karnali River fan and BNP, with the park boundaries and buffer zones indicated. The four lakes are labelled by name, and the three transects for luminescence dating are portrayed in red.

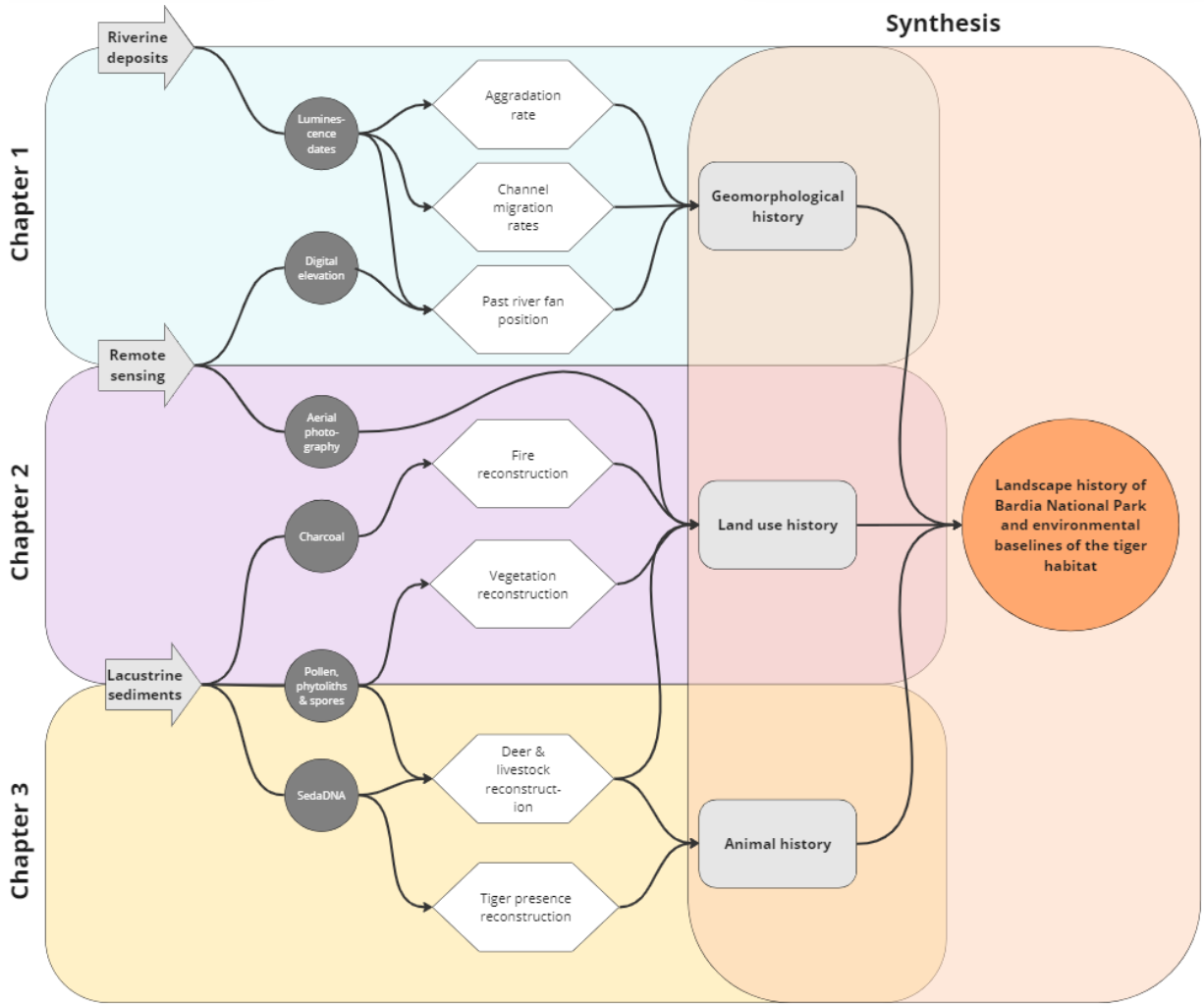

Figure 2. Schematic overview of the research structure, including the objectives and outcomes of each research chapter.

Chapter 1: Geomorphological history

Overall, two types of samples will be collected to establish the geomorphological history through optically stimulated luminescence dating. Samples can be taken from paleochannels that are visible in the banks of current river channels, in which case they are collected using 4.5 cm diameter PVC (polyvinyl chloride) tubes of ca. 30 cm long, that are hammered horizontally into the bank and then removed and sealed using black tape. Samples for the meander migration patterns and aggradation rate are collected from the subsurface through vertical auguring. A Van-der-Staaij-attachment and manual auger are used, and samples sealed with black tape.

Measuring the past channel activity in the BNP forest

To estimate the past channel activity east of- and in the center of the fan, and therefore its potential to have affected Bardia National Park, OSL-samples will be taken to date last activity of paleo channels. A small channel perpendicular to the river fan, with visible paleochannels, was selected as our main transect for establishing past channel activity east of the fan (Fig. 3A). Horizontal OSL-samples will be collected in the paleo channels to determine when they were last active. A similar reconstruction was already performed by Dingle et al. (2020), where they found that channel mobility was minimal in the northern part of the fan, having been mostly fixed at the bifurcation for 2500 years. Channels further south appear more mobile, ranging from 500 to 800 years in age. Paleochannels east of the Kauriala branch that may have originated from the fan, were dated last active 1000 – 4500 years ago. It was decided to re-use this data for our analysis if possible, and complement it with own measurements, unless we find strong evidence that further research is required.

Meander migration patterns below in the south of the river fan

In Dingle et al., 2020, a transition from gravel to sand is described about halfway down the longitudinal axis of the fan. This transition was also observed during the 2022 field visit, and satellite imagery reveals a change from a braided system above this transition, to a meandering system below. To compile a full image of the past spatial position of the river fan, the migration patterns of the meandering system below the GST (Gravel to Sand Transition) will be determined (See i.e. Quik & Wallinga, 2018). OSL-samples will be collected in the Madhuvan oxbow, along a transect running from the outer to the inner bank and back to the main active channel (Fig. 4). These samples will be obtained by coring vertically, capturing the top layer of the sand deposited by the channel in the past (Fig. 3B).

Aggradation rate

Finally, aggradation rate is considered to build understanding of temporal changes in the deposition of sediment, which can potentially be connected to specific events or transitions in climate, human occupancy and/or river morphology. Aggradation rate will be determined by collecting OSL-samples along an east-west transect along the downstream area of the fan (Fig. 4). At each coring point, several OSL-samples will be retrieved: one at the top, directly below the plough horizon (Fig. 3C), a few in between (fixed interval) and one at the bottom of the core, which end is determined by the gravel layer. From the derived ages at the top and bottom, the overall aggradation rate can be calculated, and measurements in between will be used to determine whether aggradation rate was linear over time.

Data quantification and analysis

All luminescence samples will be analyzed in the Netherlands Institute for Luminescence dating to determine exact ages. The resulting ages will be combined with analysis of a DEM (digital elevation model) (10m or 2m resolution) to develop a landscape evolution model on a Holocene timescale. It is important to note here, that under certain circumstances (i.e. extreme monsoon events), channels can abruptly shift hundreds of meters and that the reworking and erosion of sedimentary records may affect the ages that we retrieve. The Karnali river fan system is significantly more dynamic than i.e. the meandering system of the Vecht studied by Quik & Wallinga (2018). Therefore, interpreting the past channel activity in the fan has its limitations and the aim is to estimate general landscape evolution patterns rather than direct channel position reconstruction.

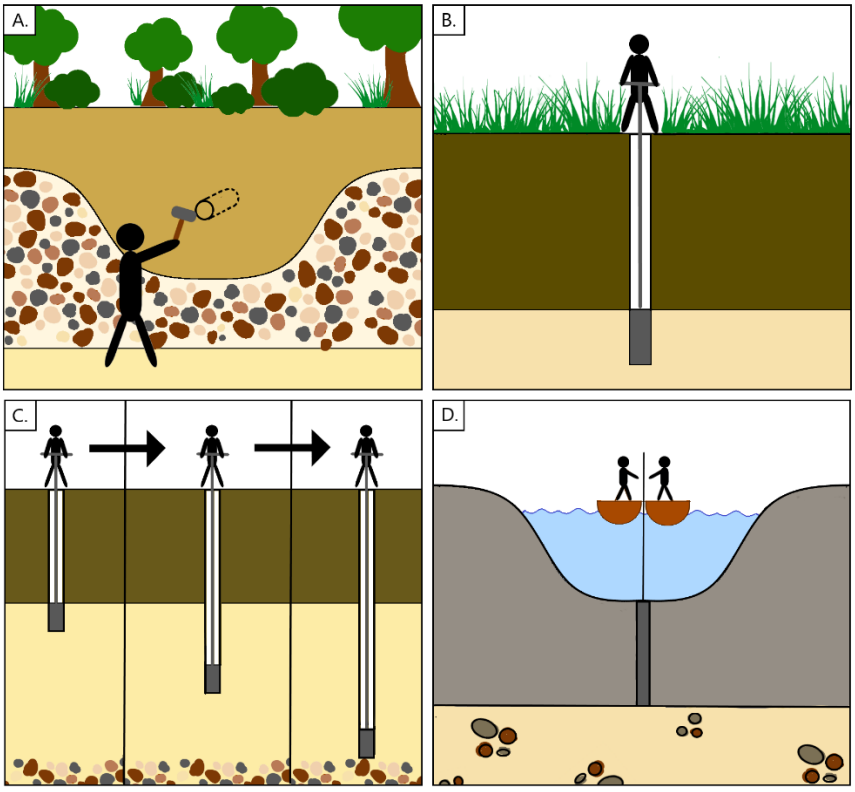

Figure 3. Schematic illustration of sampling methodologies with A) paleochannel dating along the channel transect in BNP where PVC-tubes are hammered horizontally into paleochannel exposures, B) establishing the meander migration rate along the Madhuvan oxbow transect by collecting sand deposits with a Van der Staaij attachment, C) determining the aggradation rate of the fan by sampling sand deposits at various depths with a Van der Staaij attachment and D) retrieving sediment records for paleoecological reconstruction with the Colinveaux-Vohnout corer.

Chapter 2: Reconstructing the land use history of Bardia National Park

To reconstruct the land use history, two palaeoecological record sources were considered for sampling in the BNP area: terrestrial deposits and lacustrine deposits. Upon visiting the area, however, it was quickly established that terrestrial deposits would not yield records of sufficient continuity and resolution, as both rates of surface-erosion and bioturbation by termites are very high throughout BNP. Therefore, for paleoecological reconstruction, we aim to collect sediment cores from lacustrine deposits only.

Site selection

Lakes are rare in the area of interest, but I was able to identify 4 prospective sampling sites divided over two areas of interest. One of the areas of interest is the lowland plain east of the Karnali river fan, as this is a likely area for early settlement and landscape modification by the Tharu, which would have migrated from the south. The second area of interest is the Bardia forest under direct influence of the river fan (the Geruwa branch, specifically), as this is where changes in the fan dynamics may have directly affected tiger habitat suitability, and that connects most strongly to conservation policy in BNP.

The first site considered for the lowland plains is Lake Satkhaluwa (28°19'17.05"N 81°28'18.91"E). It has a surface area of ca. 0.12 km^2 and is located in the buffer zone of BNP. A dam is present on the southern end, but satellite imagery and local interviewing established that this dam is recent, and that the lake existed before the dam was built. The lake was measured at approximately 3 meters deep during the field visit, and using a gauge we collected dark sediment from the lake bottom, indicating favourable conditions for microfossil preservation. The formation of the lake is currently unclear, so no estimation of the age can be made yet. Secondly, for the lowland plains, Lake Barhaiya is located directly south of the buffer zone (28°11'51.73"N 81°30'49.27"E) and is currently heavily influenced by anthropogenic practices such as fish farming and damming. However, oral histories support that this lake is not man made, and as it is larger than other lakes in the area (0.55 km2), the site is of interest. It is expected that the top layer of sediment is too disturbed, but a continuous record could be present below.

The first site of interest for Bardia forest lies in the Madhuvan municipality, located on the Nepal-India border (28°20'50.07"N 81°13'16.03"E). It is built atop a visible oxbow scar originating from the main Geruwa branch.

Most of the surface area on the Nepali side of the border is currently used for agricultural purposes, but a small lake area (0.07 km2) persists to this day. The age of the oxbow is unknown as of right now, but it was already present in its current form on a map from 1927 AD. Secondly, several small ponds are scattered throughout BNP itself. Although many of these are confirmed as artificial, locals informed us that some are thought to be natural. We were not able to determine which are natural in the field, but with a high-resolution digital elevation model (2m), I will search for natural depressions around these lakes.

Ideally, at least one core of each area will be analysed. Which cores are fully processed will be determined by performing preliminary tests of the preservation of pollen and phytoliths at a 30-50 cm interval, depending on the depth of the core. From Lake Satkhaluwa, we already obtained three preliminary samples with a gouche that are currently being processed to determine phytolith concentration, to gain the first insight on microfossil influx and preservation. Currently, the ages of our sites are unknown, but a conceptual landscape evolution model that will be developed by a MSc-student, will be also be used to consider which lakes are most likely to be of sufficient age. The conceptual model will later be verified by generating age-depth models from radiocarbon dates for each of the cores before performing full analyses.

Data collection and analysis

To collect continuous sediment cores, a Colinveaux-Vohnout coring rig will be used (Fig 3D). At each site, one sediment core will be retrieved for pollen, spores, phytoliths & charcoal analysis. As the Colinveaux-Vohnout coring equipment is specifically designed for soft and fine sediments, the depth of the core will be limited to where sand, gravel or rock is reached. If sand is found to be the limiting material, it will be attempted to retrieve a small section of sand to apply luminescence dating, which can provide a preliminary insight on the temporal scale of the record.

The sediment cores will be subsampled at 1 cm intervals. Half of each subsample will remain in Nepal at the department of Geology at Tribhuvan University to perform pollen analysis by protocol of Wood et al. (1996) under supervision of professor Khum Paudayal, which will result in a vegetation record on a regional scale. The other half will be taken to Wageningen, where 3 cm2 will be prepared for phytolith analysis per protocol by Piperno (2006) to provide local vegetation history and charcoal will be extracted to establish fire history. The resulting assemblages of both phytoliths and pollen will be run through CONISS analysis to statistically determine the changes over time, where the focus lies on identifying transitions between riverine forest, sal forest, mixed forests, phantas and riverine grasslands. To aid the interpretation of these transitions, the generated fire records, and available literature of archaeology (i.e. Corvinus, 2007) will be utilized. In addition, phytolith data will be run through a Detrended Correspondence Analysis (DCA) to identify whether differences between specifically riverine grasslands and phantas can be identified from the phytolith assemblage alone (i.e. Witteveen et al., 2022). If it is possible to make a distinguishment, this will increase understanding of the development of both grassland types over time.

Distinguishing different drivers through time will mainly be done by making optimal use of the multi-proxy approach described above. Each of the proposed drivers has indicators that are independent of the vegetation and animal responses, as river morphology will be measured with luminescence dating and remote sensing, human activity is also indicated by archaeological findings and global migration patterns, and general climate patterns can be observed from records from Lake Rara. Aligning all of these factors with the response in vegetation and animal occupancy will allow us to reason and interpret the drivers of the identified transitions over time.

Chapter 3: Reconstructing the animal history of Bardia National Park

The reconstruction of animal history will be based on sediment cores collected at the same sites as chapter 2. Microscopy slides prepared for pollen analysis in the context of chapter 2 also contain non-pollen palynomorphs (NPP's) such as spores of coprophilous fungal (SCF). Therefore, after pollen analysis at Tribhuvan University, the

microscopy slides will be shipped to Wageningen University, where SCF counts will be performed to estimate the herbivore density over time. This density will be used to inform livestock density, but in the absence of any other signals of human activity, the counts can be used to estimate wild herbivore density which is relevant to the tiger habitat suitability regarding prey availability. A second sediment core designated to sedaDNA analysis will be retrieved from each lake site. The same general coring methodology is to be applied, with the addition of thoroughly cleaning the coring tubes with bleach before use. The cores will be subsampled at 1 cm intervals in a sterile environment, after which samples will be analysed using shotgun sequencing methodology (Pedersen et al., 2016) to identify where tiger DNA may be present. Positive outcomes will be verified by targeted quantitative analysis targeting the continental tiger. Using the resulting record, a presence-absence reconstruction over time will be made.

The presence-absence reconstruction of the tiger, the estimate of herbivore density and the indicators for human activity measured in chapter 2 will be integrated into an understanding of the animal history of Bardia National Park. This will then feed into the synthesis of a final, full reconstruction of the habitat of the continental tiger over the past millennia and potential environmental baselines that can be used to develop conservation & management plans of the area.

Timeline

The PhD-project is scheduled from September 1st, 2022, to September 1st, 2026. The Gantt-chart in figure X shows the intended planning of the PhD-project in monthly intervals. Graduate school-required tasks are marked in red, fieldwork in yellow, laboratory & analysis work in purple and writing in blue. Each chapter analysis is followed by a period dedicated to writing and submitting a corresponding paper. It is foreseen that chapter 1 will result in a paper on changes in river morphology over time, possibly in collaboration with work package 4 at TU Delft. Chapter 2 is planned to generate a paper on distinguishing phantas from riverine grasslands in phytolith assemblages. Chapter 3 will lead to a publication on obtaining temporal mammalian presence data from subtropical sediment records through sedaDNA analysis and finally a fourth paper will be written that integrates all outcomes from the three chapters to provide an elaborate, multidisciplinary landscape reconstruction of the tiger habitat in the Terai Arc Landscape.

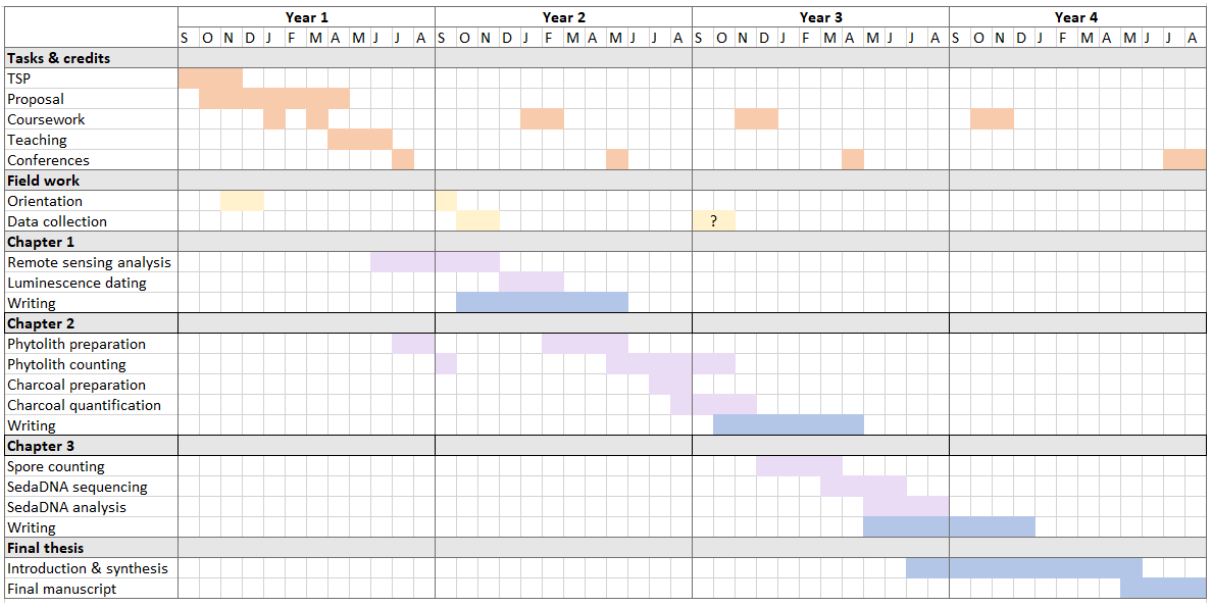

Figure 4. Timeline of the full PhD-project portrayed in a Gantt-chart.

Feasibility

The funding for this project is sufficient (Tbl. 1). Several risks for data collection and retrieval have been identified during the first field visit. The scarcity of lakes provides a risk in the quality and continuity of sedimentary records

used for the paleoecological reconstruction. This risk is however largely mitigated by taking sediment cores from 4 different lakes that appear to have different origins, increasing the chances that at least one retrieval is successful. In addition, preliminary samples from one of these lakes are currently being prepared for phytolith extraction to get an idea of the preservation quality. Further methodological risks also lie in the highly experimental nature of sedaDNA analysis and luminescence dating. The foundations of this project, however, do not rely on sedaDNA analysis alone as traditional paleoecological proxies will also be applied, providing a reliable supporting data set. In regards to luminescence dating, it is promising that Dingle et al. (2020) were able to generate dates from their samples, but if using quartz grains does not yield results, it is possible to try a different approach using feldspar grains instead. Finally, the geomorphological reconstruction can also be supported with remote sensing methods and landscape evolution modelling.

Table 1. Short overview of project income & costs

Incoming	€/unit	Qty.	Total
NWA funding	65,140	1	65,140
PE&RC	3,500	1	3,500
			68,640
Costs			
Fieldwork	7,500	2	15,000
Luminescence dating	700	25	17,500
Radiocarbon dating	500	15	7,500
Coring equipment	5000	1	5,000
Other/extra analyses	500	20	12,140
Training	875	4	3,500
Conferences	2,000	4	8,000
			68,640

2.6 Data management Data generated in this project will be stored on OneDrive and SharePoint during the research phase. From there, data is only accessible to collaborators. Upon publication of articles and/or the manuscript, data will be made accessible through the Dutch repository EASY under DANS (Data Archiving and Networked services). The data is owned by Wageningen University & Research. Full details on data management are available in the data management plan that was submitted to the Graduate school for Production Ecology & Resource Conservation in March 2023.

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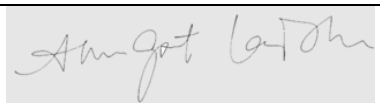
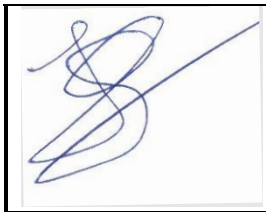
Part 3: Reviewers and signatures

3.1 POTENTIAL REVIEWERS

The proposal will be sent to 3 reviewers. Please provide the names and contact details of 4-5 potential reviewers who are in no way involved in the project. A balanced representation of gender and reviewers from inside and outside the main PE&RC affiliated institutes are preferred. To speed up the review process, it is allowed and even encouraged to verify the proposed reviewer's willingness to provide an independent review of the proposal.

#	Name + title	Organisation	Specialisation	Email address
1.	Dr. Elizabeth Dingle	Durham University	Fluvial geomorphology	Elizabeth.dingle@durham.ac.uk
2.	Professor Frank van Langevelde	Wageningen University	Conservation biology	Frank.vanlangevelde@wur.nl
3.	Dr. Carina Hoorn	University of Amsterdam	Paleoecology	m.c.hoorn@uva.nl
4.	Dr. Jan-Henrik May	University of Melbourne	Fluvial and Mega Fan Geomorphology	janhendrikmay@unimelb.edu.au
5.	Professor Inger. G. Alsos	The Arctic University of Norway	SedaDNA	Inger.g.alsos@uit.no

3.2 SIGNATURES

PhD candidate	**Principal supervisor / Promotor**	**Supervisor 2**
Name: Zoë Sophia Kleijwegt	Name: Prof. Dr. Jakob Wallinga	Name: Dr. Annegret Larsen
Date: 25/08/2023	Date: 25/08/2023	Date:25.8.2023
		
Supervisor 3	**Supervisor 4**	**Supervisor 5**
Name: Dr. Jeroen M. Schoorl	Name:	Name:
Date:25-08-2023	Date:	Date: